

PREMIER UNIVERSITY

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E-Commerce Sales Data Analysis

Using Hadoop, MapReduce and Pig

Big Data Analytics

Project Report

Submitted To

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1. Introduction

The rapid growth of e-commerce has resulted in the generation of large amounts of transactional data. This data contains valuable information about products, customers, quantities, prices, countries, and sales revenue.

Big Data technologies provide scalable methods for storing, processing, and analysing such datasets. Hadoop provides a distributed environment where large datasets can be stored and processed efficiently.

In this project, an e-commerce sales dataset is analysed using three major Big Data technologies:

- Hadoop Distributed File System (HDFS) for data storage.
- Hadoop MapReduce for data processing.
- Apache Pig for data analysis.

The complete workflow demonstrates how a raw e-commerce dataset can be stored, processed, and analysed using Big Data technologies.

2. Objectives

The main objectives of this project are:

1. To store an e-commerce dataset using HDFS.
2. To process sales data using Hadoop MapReduce.
3. To analyse sales data using Apache Pig.
4. To calculate total sales revenue.
5. To calculate total quantity sold.
6. To determine country-wise sales revenue.
7. To determine country-wise quantity sold.
8. To calculate average unit price by country.
9. To demonstrate a practical Big Data analytics workflow.

3. Dataset

The dataset used in this project is the **Online Retail Dataset**. It contains transactional records from an online retail business.

The major attributes used in the project are:

Table 1: Dataset Attributes

Attribute	Description
InvoiceNo	Invoice number
StockCode	Product stock code
Description	Product description
Quantity	Quantity of products purchased
InvoiceDate	Date and time of transaction
UnitPrice	Price of a single product
CustomerID	Customer identifier
Country	Customer country
TotalPrice	Total transaction value

The `TotalPrice` field was used for revenue-related calculations.

4. Tools and Technologies

The following technologies were used:

Table 2: Tools and Technologies

Technology	Purpose
Hadoop	Big Data processing framework
HDFS	Distributed data storage
MapReduce	Data processing
Apache Pig	Data analysis
Java	MapReduce programming language
Windows CMD	Command-line execution
GitHub	Project version control and hosting
Overleaf	Report preparation

5. Project Workflow

The overall project workflow consists of three major stages:

1. Store the dataset in HDFS.
2. Process the dataset using Hadoop MapReduce.
3. Analyse the dataset using Apache Pig.

The simplified workflow is:

Online Retail Dataset → HDFS → MapReduce → Pig Analysis → Results

6. HDFS Data Storage

Hadoop Distributed File System (HDFS) was used to store the cleaned e-commerce dataset.

The dataset was uploaded to the following HDFS location:

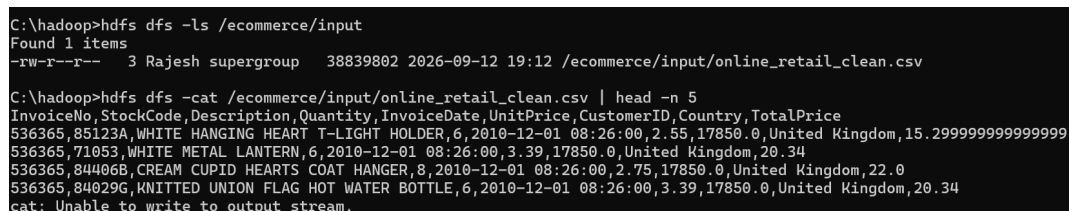
```
/ecommerce/input/online_retail_clean.csv
```

The following commands were used to verify the dataset:

```
hdfs dfs -ls /ecommerce/input
```

```
hdfs dfs -cat /ecommerce/input/online_retail_clean.csv | head -n 5
```

The first command displays the dataset stored in the HDFS directory, while the second command displays the first few records of the dataset.



```
C:\hadoop>hdfs dfs -ls /ecommerce/input
Found 1 items
-rw-r--r--  3 Rajesh supergroup  38839802 2026-09-12 19:12 /ecommerce/input/online_retail_clean.csv

C:\hadoop>hdfs dfs -cat /ecommerce/input/online_retail_clean.csv | head -n 5
InvoiceNo,StockCode,Description,Quantity,InvoiceDate,UnitPrice,CustomerID,Country,TotalPrice
536365,85123A,WHITE HANGING HEART T-LIGHT HOLDER,6,2010-12-01 08:26:00,2.55,17850.0,United Kingdom,15.299999999999999
536365,71053,WHITE METAL LANTERN,6,2010-12-01 08:26:00,3.39,17850.0,United Kingdom,20.34
536365,84406B,CREAM CUPID HEARTS COAT HANGER,8,2010-12-01 08:26:00,2.75,17850.0,United Kingdom,22.0
536365,84029G,KNITTED UNION FLAG HOT WATER BOTTLE,6,2010-12-01 08:26:00,3.39,17850.0,United Kingdom,20.34
cat: Unable to write to output stream.
```

Figure 1: Storing and accessing E-Commerce dataset using HDFS

Figure 1 shows the successful storage and access of the e-commerce dataset through HDFS.

7. MapReduce Data Processing

Hadoop MapReduce was used to process the e-commerce dataset and calculate revenue by country.

The MapReduce program consists of two main components:

- **Mapper:** Reads each transaction and emits the country as the key and transaction revenue as the value.
- **Reducer:** Groups records by country and calculates the total revenue for each country.

The MapReduce program was executed using:

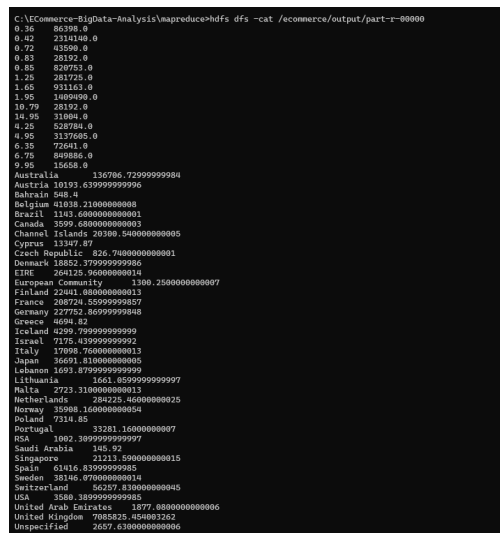
```
hadoop jar ECommerceRevenue.jar ECommerceRevenue \
/ecommerce/input /ecommerce/output
```

The MapReduce job completed successfully with the following important statistics:

- Map input records: 392,693
- Map output records: 390,391
- Reduce input groups: 52
- Reduce output records: 52
- Map completion: 100%
- Reduce completion: 100%

The output was viewed using:

```
hdfs dfs -cat /ecommerce/output/part-r-00000 | head -n 15
```



```
C:\ECommerce-BigData-Analysis\mapreduce>hdfs dfs -cat /ecommerce/output/part-r-00000
0.36 86398.0
0.42 2314148.0
0.72 43528.0
0.83 26192.0
0.85 829753.0
1.25 261725.0
1.65 931163.0
1.95 1808894.0
18.79 26192.0
18.95 31864.0
4.25 526784.0
4.95 3137685.0
6.35 72641.0
6.75 849886.0
9.95 15658.0
Australia 136786.729999999984
Austria 10193.639999999996
Bahrain 548.4
Belgium 41038.210000000008
Brazil 1193.6800000000001
Canada 3599.6800000000003
Channel Islands 26380.540000000005
Cyprus 133497.87
Czech Republic 826.7400000000001
Denmark 18852.779999999996
EIME 264125.160000000014
European Community 1380.2500000000007
Farland 22481.880000000013
France 208724.559999999837
Germany 227752.869999999848
Greece 4694.82
Iceland 4299.7999999999999
Israel 7175.4399999999992
Italy 17068.760000000013
Japan 36691.810000000005
Lebanon 1693.8799999999999
Lithuania 1661.8599999999997
Malta 2723.3100000000013
Netherlands 284225.860000000025
Norway 35988.1600000000054
Poland 7314.82
Portugal 33281.160000000007
RSA 1062.3899999999997
Saudi Arabia 145.52
Singapore 21213.5900000000015
Spain 83116.839999999995
Sweden 38146.8700000000014
Switzerland 56257.8300000000045
USA 3580.3899999999995
United Arab Emirates 1877.8800000000006
United Kingdom 708325.8500000000262
Unspecified 2657.6300000000006
```

Figure 2: Country-wise revenue processing using Hadoop MapReduce

Figure 2 shows the successful execution and output of the MapReduce processing stage.

8. Apache Pig Data Analysis

Apache Pig was used to perform higher-level analysis of the e-commerce sales dataset.

The Pig script loaded the dataset from HDFS and performed several operations, including grouping, filtering, summation, and average calculations.

The following analyses were performed:

- Total revenue.
- Total quantity sold.
- Country-wise revenue.
- Country-wise quantity.
- Country-wise average unit price.

The Pig script was executed using:

```
pig -x mapreduce ecommerce_analysis.pig
```

The analysis generated the following HDFS output directories:

```
/ecommerce/pig/total_revenue
/ecommerce/pig/total_quantity
/ecommerce/pig/country_revenue
/ecommerce/pig/country_quantity
/ecommerce/pig/country_avg_price
```

The Pig execution completed successfully.

```
C:\ECommerce-BigData-Analysis\pig>hdfs dfs -cat /ecommerce/pig/total_revenue/part-r-00000
8679794.134002768

C:\ECommerce-BigData-Analysis\pig>hdfs dfs -cat /ecommerce/pig/total_quantity/part-r-00000
5032212

C:\ECommerce-BigData-Analysis\pig>hdfs dfs -cat /ecommerce/pig/country_revenue/part-r-00000 | head -
n 15
RSA,1002.3099999999998
USA,3580.3900000000002
EIRE,264125.95999999995
Italy,17098.7600000000013
Japan,36691.809999999997
Malta,2723.3100000000001
Spain,61416.839999999996
Brazil,1143.6000000000001
Canada,3599.6800000000003
Cyprus,13347.8699999999983
France,208724.559999999837
Greece,4694.8199999999999
Israel,7175.4399999999998
Norway,35908.1600000000054
Poland,7314.8499999999985
```

Figure 3: E-Commerce sales data analysis using Apache Pig

Figure 3 shows the output generated after executing the Pig analysis.

9. Results

The analysis produced the following major results.

Table 3: Overall Analysis Results

Metric	Result
Total Revenue	£8,679,794.13
Total Quantity Sold	5,032,212
Highest Revenue Country	United Kingdom
UK Revenue	£7,085,825.45
UK Quantity Sold	4,128,898

The total revenue generated by the transactions was approximately **£8.68 million**. The total quantity sold was **5,032,212 units**.

The United Kingdom generated the highest revenue in the analysed dataset, with approximately **£7.09 million** in revenue.

10. Implementation Summary

The project successfully demonstrated the three required Big Data operations.

10.1 HDFS Storage Operation

HDFS was used to store the cleaned Online Retail dataset. The dataset was successfully uploaded to the HDFS input directory and verified using HDFS commands.

10.2 MapReduce Processing Operation

MapReduce was used to process the transaction records and calculate country-wise revenue. The Mapper generated country and revenue pairs, while the Reducer aggregated revenue values by country.

10.3 Pig Analysis Operation

Apache Pig was used to perform analytical operations on the dataset. Grouping, filtering, summation, and average functions were used to obtain useful business insights.

11. Advantages

The project demonstrates several advantages of using Big Data technologies:

- Efficient storage using HDFS.
- Scalable processing using MapReduce.
- Simple data analysis using Pig Latin.
- Ability to process large transactional datasets.
- Separation of storage, processing, and analysis operations.
- Reproducible command-line based workflow.

12. Limitations

There are some limitations in the current implementation:

- The project uses a single-machine Hadoop environment.
- The MapReduce implementation uses a simple CSV parsing approach.
- Advanced visualization is not included.
- Real-time data processing is not implemented.
- The dataset represents historical transactions rather than real-time e-commerce activity.

13. Future Improvements

The project can be extended in several ways:

- Use Apache Spark for faster large-scale processing.
- Use Hive for SQL-based Big Data analysis.
- Develop dashboards using Power BI or Tableau.
- Implement product-level sales analysis.
- Implement customer segmentation.
- Perform monthly and yearly sales trend analysis.
- Develop predictive sales models.
- Deploy the project on a multi-node Hadoop cluster.

14. Conclusion

This project successfully demonstrated an end-to-end Big Data analytics workflow for e-commerce sales data using Hadoop, MapReduce, and Apache Pig.

HDFS was used to store the dataset, MapReduce was used to process the transaction records, and Apache Pig was used to perform analytical operations.

The analysis identified a total revenue of approximately **£8,679,794.13** and a total quantity sold of **5,032,212 units**. The United Kingdom was identified as the highest-revenue country in the analysed data.

Overall, the project provided practical experience with Hadoop-based Big Data storage, processing, and analysis and demonstrated how these technologies can be applied to real-world e-commerce data.

15. GitHub Repository

The complete project source code, dataset-related files, MapReduce program, Pig script, and screenshots are available in the GitHub repository:

<https://github.com/RajeshDas57/ECommerce-BigData-Analysis>

The repository contains:

- Dataset files.
- Hadoop MapReduce source code.
- MapReduce JAR file.
- Apache Pig script.
- Project screenshots.

16. References

1. Apache Hadoop Documentation.
2. Apache Pig Documentation.
3. Hadoop: The Definitive Guide.
4. Online Retail Dataset.
5. Apache Hadoop MapReduce Documentation.